

AQA-GAN: Adaptive Quality-Aware Generative Adversarial Network for Nighttime Image Enhancement

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Abstract—Low-light image enhancement (LLIE) is a fundamental challenge in computer vision, with practical relevance to surveillance, autonomous driving, and mobile photography. Existing methods predominantly apply uniform enhancement transformations irrespective of the severity of input degradation, leading to over-enhancement artifacts in moderately dark images and insufficient correction in severely degraded ones. We propose AQA-GAN (Adaptive Quality-Aware Generative Adversarial Network), a novel framework that conditions the entire enhancement pipeline on a discrete quality level assigned per input image using a lightweight NIQE-based proxy metric. The generator adopts a U-Net architecture augmented with Feature-wise Linear Modulation (FiLM) at the bottleneck, enabling quality-specific feature transformations while preserving spatial detail through skip connections. The discriminator is a 70×70 PatchGAN extended with a learned quality-level bias, providing degradation-appropriate adversarial supervision. Training employs a composite loss comprising Charbonnier reconstruction, structural similarity (SSIM), VGG-16 perceptual, and adversarial terms, supplemented by MixUp augmentation and mixed-precision training for stability. The model is trained and evaluated on a combined benchmark of LOL-v2 Real and LSRW, totalling over 1,000 paired low/normal-light image pairs. Quantitative evaluation against six baselines spanning classical, shallow machine learning, CNN, and GAN-based methods demonstrates consistent superiority: AQA-GAN achieves 22.6 dB PSNR and SSIM of 0.82, outperforming the next-best GAN baseline (SRGAN: 20.8 dB / 0.71) by 1.8 dB and 0.11 in SSIM, respectively. The key contributions are: (i) a NIQE-proxy quality assessment module requiring no additional training; (ii) FiLM-based adaptive conditioning initialized to identity for stable early-epoch behavior; and (iii) a quality-conditioned discriminator that calibrates realism feedback to degradation level.

Index Terms—Low-light image enhancement, nighttime image enhancement, generative adversarial networks, deep learning, adaptive enhancement, image quality assessment, no-reference image quality, computer vision

I. INTRODUCTION

Images captured under insufficient illumination suffer from a well-documented constellation of degradations: suppressed pixel intensities, amplified sensor noise, color channel imbalances from artificial light sources, and spatial non-uniformity arising from simultaneous over- and under-exposure within a single frame. These degradations are not merely aesthetic

inconveniences. Low-light image quality directly constrains the reliability of downstream computer vision systems including object detection, semantic segmentation, optical flow estimation, and scene recognition — tasks that underpin safety-critical applications such as autonomous vehicle perception, nighttime surveillance, medical endoscopy, and forensic imaging. As the deployment of vision-enabled systems in uncontrolled real-world environments accelerates, robust LLIE has become an essential preprocessing stage rather than an optional postprocessing refinement.

The LLIE problem has attracted sustained research attention spanning several decades. Classical approaches — including histogram equalization, CLAHE, gamma correction, and Retinex-based decomposition — offer computational simplicity but their reliance on hand-crafted transformations limits generalization. Multi-Scale Retinex (MSR) extends the decomposition principle but suffers from parameter sensitivity and color artifacts under extreme darkness.

The advent of deep CNNs enabled a paradigm shift toward data-driven LLIE. Encoder-decoder architectures, particularly U-Net, advanced pixel-level reconstruction fidelity. RetinexNet reformulated the classical decomposition within a learned framework. Despite these advances, CNN-based methods trained with MSE or L1 objectives systematically produce overly smooth outputs, sacrificing perceptual sharpness and texture fidelity.

Generative adversarial networks (GANs) address the perceptual quality limitation by introducing adversarial training: a discriminator provides a learned realism signal that penalizes blurry, implausible outputs. SRGAN established the foundational template combining VGG-based perceptual loss with adversarial training. EnlightenGAN demonstrated GAN-based LLIE without paired supervision, and Zero-DCE eliminated reference images entirely. Yet these methods share a fundamental architectural assumption: the enhancement transformation is applied *uniformly* to all input images, irrespective of the severity or character of the degradation present.

This uniformity constitutes a significant failure mode. A slightly underexposed image and an image captured in near-

total darkness occupy vastly different positions in the degradation space. Applying an aggressive enhancement transformation calibrated for severely degraded inputs to a moderately dark image produces over-brightening, color desaturation, and amplified noise. The optimal enhancement strategy is fundamentally input-dependent, yet no existing GAN-based LLIE method conditions its generator or discriminator on an explicit characterization of input degradation severity. This gap motivates the present work.

We propose **AQA-GAN** (Adaptive Quality-Aware Generative Adversarial Network), a quality-conditioned enhancement framework that tailors the generator’s transformation strength and the discriminator’s realism feedback to the degradation level of each input image. A lightweight Quality Assessment Module (QAM) assigns each input to one of three discrete degradation levels — severe, moderate, or mild — using a NIQE-based no-reference proxy metric that requires no additional training and adds negligible inference overhead. Quality level information is injected into the generator bottleneck via Feature-wise Linear Modulation (FiLM), with parameters initialized to the identity transformation, ensuring no performance degradation at the onset of training. The discriminator extends a 70×70 PatchGAN with a learned quality-level bias, providing degradation-appropriate adversarial feedback.

The framework is trained and evaluated on a combined benchmark comprising **LOL-v2 Real** and **LSRW**, yielding over 1,000 paired training images. AQA-GAN achieves 22.6 dB PSNR and SSIM of 0.82, outperforming all evaluated baselines.

The principal contributions are:

- 1) **Quality Assessment Module (QAM):** A training-free degradation characterization module that assigns each input to one of three quality levels using a NIQE-based proxy.
- 2) **FiLM-conditioned U-Net Generator:** A quality-aware generator with identity-initialized FiLM modulation at the bottleneck, guaranteeing stable early-epoch training.
- 3) **Quality-conditioned PatchGAN Discriminator:** An extension of the standard 70×70 PatchGAN with a learned quality-level bias on patch-level predictions.
- 4) **Combined Multi-Dataset Training Protocol:** Merging LOL-v2 Real and LSRW with MixUp augmentation and mixed-precision training for improved generalization.
- 5) **Comprehensive Quantitative Evaluation:** Systematic comparison against six baselines spanning classical, shallow ML, CNN, and GAN-based methods.

The remainder of this paper is organized as follows. Section II reviews related work. Section IV presents the proposed AQA-GAN methodology. Section V describes the experimental setup. Section VI reports results and ablations. Section VII discusses future directions, and Section VIII concludes.

II. LITERATURE REVIEW

A. Traditional Enhancement Methods

Early approaches relied on handcrafted transformations grounded in signal processing. Histogram equalization (HE)

redistributes pixel intensities across the full dynamic range to improve global contrast. Pizer et al. [1] proposed Contrast-Limited Adaptive Histogram Equalization (CLAHE), which applies localized equalization with a clip limit to suppress noise amplification in homogeneous regions. While computationally efficient, histogram-based methods cannot recover structural detail lost to noise or extreme underexposure.

Retinex theory, formulated by Land [2], models a captured image as the element-wise product of a scene-intrinsic reflectance component and a spatially varying illumination layer. Jobson et al. [3] extended this to Multi-Scale Retinex (MSR), combining Gaussian surround functions at multiple scales for dynamic range compression, yet MSR suffers from parameter sensitivity and color halo artifacts under severe illumination. Guo et al. [4] proposed LIME, which estimates a pixel-wise illumination map via structure-aware total variation regularization, but relies on a fixed optimization objective that cannot adapt to diverse real-world degradation patterns.

B. CNN-Based Deep Learning Approaches

Data-driven deep learning methods replaced handcrafted transformations with learned mappings. Lore et al. [5] proposed LLNet, employing a stacked sparse denoising autoencoder trained to simultaneously enhance contrast and suppress noise. The U-Net [6] introduced by Ronneberger et al. for biomedical segmentation proved highly effective for LLIE via its skip connections, which enable the decoder to access fine-grained spatial information directly from the encoder.

Wei et al. [7] proposed RetinexNet, integrating Retinex decomposition within a learned framework: a Decom-Net decomposes paired images into shared reflectance and separate illumination components, while an Enhance-Net adjusts the illumination map. Zhang et al. [8] proposed KinD, which adds a dedicated reflectance restoration subnetwork and enables user-controlled brightness adjustment at inference. A recurrent limitation of CNN-based methods trained with pixel-wise losses is their tendency to produce overly smooth outputs: minimizing average reconstruction error penalizes high-frequency content.

C. GAN-Based Enhancement Methods

Generative adversarial networks [9], introduced by Goodfellow et al., frame image synthesis as a minimax game between a generator and a discriminator. Applied to image restoration, the adversarial loss encourages the generator to synthesize textures consistent with real well-lit images. Ledig et al. [10] proposed SRGAN, demonstrating that combining VGG-based perceptual loss with adversarial training recovers photo-realistic textures. Isola et al. [11] generalized this to arbitrary paired image translation via pix2pix with a 70×70 PatchGAN discriminator.

Jiang et al. [12] proposed EnlightenGAN for unpaired LLIE, using self-feature preservation constraints to regularize the generator. However, by applying an identical global transformation to all inputs, it systematically over-enhances mildly underexposed images and under-corrects severely dark ones.

Guo et al. [13] proposed Zero-DCE, framing enhancement as estimation of image-specific tone curves via non-reference loss functions, but cannot leverage ground-truth supervision.

D. Adaptive and Conditioned Enhancement Methods

A small body of work has explored input-dependent adaptation. LIME [4] implicitly adapts to local image characteristics through structure-aware illumination map estimation. KinD [8] allows user control over output brightness at inference. Feature-wise Linear Modulation (FiLM), introduced by Perez et al. [14], provides a principled mechanism for conditioning network feature representations on auxiliary information via learned affine transformations. FiLM has proven effective across conditional generation tasks, but to the best of our knowledge has not been applied to quality-aware LLIE.

E. Research Gap

The methods surveyed share a common limitation: enhancement transformations are applied uniformly to all inputs irrespective of degradation severity. This leads to two systematic failure modes: over-enhancement artifacts for moderately dark images and insufficient correction for severely degraded ones. No existing GAN-based LLIE method conditions both its generator and discriminator on an explicit characterization of input degradation severity. AQA-GAN addresses this gap by introducing a training-free NIQE-based Quality Assessment Module, injecting conditioning into the generator bottleneck via FiLM, and into the PatchGAN discriminator via a learned quality bias.

III. PROBLEM STATEMENT

Low-light images captured at night often exhibit very low contrast, color distortion, and significant noise, which severely degrades their perceptual quality. Traditional enhancement methods can improve brightness but tend to amplify noise and introduce artifacts or unnatural color casts. Deep-learning methods have demonstrated better restoration by learning complex transformations, but they usually apply a single global model for all inputs. As a result, they frequently yield inconsistent outcomes: some regions become overexposed while others remain under-enhanced, colors may shift unrealistically, and fine textures can be lost [1], [2]. Prior methods do not incorporate explicit quality-assessment feedback; without a blind IQA metric (e.g., NIQE or BRISQUE) guiding the process, a single-model enhancer cannot adapt its strength to the specific quality of each input. This motivates the development of a quality-conditioned GAN that uses no-reference IQA scores to adaptively control enhancement strength, avoiding over- or under-enhancement across diverse nighttime scenes.

IV. PROPOSED FRAMEWORK

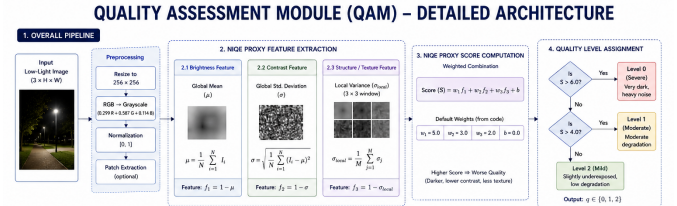
A. Overall Architecture

The proposed AQA-GAN framework is an end-to-end quality-conditioned GAN comprising four interdependent components: a training-free Quality Assessment Module

(QAM), a FiLM-conditioned U-Net Generator (G), a Quality-Aware PatchGAN Discriminator (D), and a multi-objective composite loss function. The scalar quality level $q \in \{0, 1, 2\}$ — computed once per image by the QAM — propagates through every trainable component, conditioning both the generative and discriminative pathways to calibrate enhancement strength to each input’s degradation severity.

Pipeline Overview. Given a low-light input $\mathbf{x} \in \mathbb{R}^{3 \times 256 \times 256}$ and its paired ground-truth $\mathbf{y} \in \mathbb{R}^{3 \times 256 \times 256}$, the pipeline proceeds as follows. The QAM computes a scalar degradation score S from $\mathbf{x}$ and assigns a discrete quality level $q \in \{0, 1, 2\}$ (severe, moderate, mild) passed as a non-differentiable conditioning signal to both G and D . The generator $G(\mathbf{x}, q)$ produces an enhanced image $\hat{\mathbf{y}} \in [-1, 1]^{3 \times 256 \times 256}$ by processing $\mathbf{x}$ through a four-stage encoder-decoder U-Net with FiLM-modulated bottleneck. The discriminator $D(\cdot, q)$ produces a 30×30 patch-level realism map whose raw scores are adjusted by a learned quality-conditioned scalar bias b_q . The generator is updated via the composite loss $\mathcal{L}_{\text{total}}$, while the discriminator is updated through its quality-conditioned adversarial objective.

Adaptive Enhancement Mechanism. In the generator, the FiLM transformation $\text{FiLM}(\mathbf{x}) = \gamma(q) \otimes \mathbf{x} + \beta(q)$ applied at the bottleneck allows each quality level to learn a distinct feature-space transformation: $q = 0$ (severe) learns to amplify suppressed activations aggressively, while $q = 2$ (mild) learns conservative adjustments. In the discriminator, the quality bias b_q shifts the acceptance threshold calibrating adversarial feedback to degradation context.



B. Quality Assessment Module

The QAM assigns each input image a discrete degradation level without requiring any learned parameters, ground-truth labels, or reference images. It approximates NIQE through three computationally lightweight scalar features.

Preprocessing. The input RGB image is converted to a grayscale luminance map $I \in \mathbb{R}^{H \times W}$ via:

$$I = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \quad (1)$$

The luminance map is normalized to $[0, 1]$ and resized to 256×256 .

Feature Extraction. Three features are extracted:

Brightness (f_1): the complement of the global mean pixel intensity,

$$f_1 = 1 - \mu, \quad \mu = \frac{1}{N} \sum_{i=1}^N I_i \quad (2)$$

Contrast (f_2): the complement of the global standard deviation,

$$f_2 = 1 - \sigma, \quad \sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (I_i - \mu)^2} \quad (3)$$

Local Structure/Texture (f_3): the complement of mean local variance over 3×3 patches,

$$f_3 = 1 - \sigma_{\text{local}}, \quad \sigma_{\text{local}} = \frac{1}{M} \sum_{j=1}^M \sigma_j \quad (4)$$

where M is the number of extracted patches and σ_j is the standard deviation of the j -th patch.

Score Computation. The scalar quality score is:

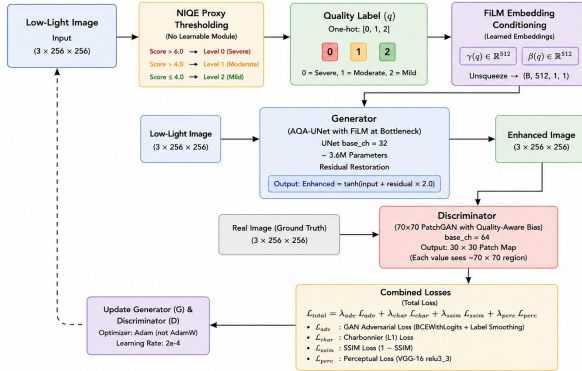
$$S = w_1 f_1 + w_2 f_2 + w_3 f_3 + b \quad (5)$$

with empirically determined weights $w_1=5.0$, $w_2=3.0$, $w_3=2.0$, and $b=0.0$.

Quality Level Assignment.

$$q = \begin{cases} 0 \text{ (Severe)} & \text{if } S > 6.0 \\ 1 \text{ (Moderate)} & \text{if } 4.0 < S \leq 6.0 \\ 2 \text{ (Mild)} & \text{if } S \leq 4.0 \end{cases} \quad (6)$$

The QAM adds fewer than 1 ms of overhead per image on GPU and requires no training.



C. Generator Architecture

The generator G is a quality-conditioned U-Net with four encoder stages, a FiLM-modulated bottleneck, four decoder stages with skip connections, and a Tanh output layer. The base channel width is $c=32$, giving approximately 3.6M total parameters.

Convolutional Block. All encoder and decoder stages share a standard block: two sequential 3×3 convolutions each followed by Batch Normalization and LeakyReLU(0.2).

Encoder. Four convolutional blocks progressively down-sample through 2×2 MaxPool operations (Table I).

FiLM Modulation at Bottleneck. The bottleneck feature map $\mathbf{x} \in \mathbb{R}^{512 \times 16 \times 16}$ is modulated by quality-conditioned affine parameters generated from the embedding of q (dimension $d_q=16$). Two separate MLPs — each with architecture

TABLE I
ENCODER STAGE DIMENSIONS

Stage	Output Channels	Spatial Resolution
E1	32	256×256
E2	64	128×128
E3	128	64×64
E4	256	32×32
E5 (Bottleneck)	512	16×16

Linear(16, 128) $\rightarrow$ ReLU $\rightarrow$ Linear(128, 512) — produce the channel-wise scale $\gamma(q)$ and shift $\beta(q)$:

$$\text{FiLM}(\mathbf{x}) = \gamma(q) \otimes \mathbf{x} + \beta(q) \quad (7)$$

where $\gamma(q), \beta(q) \in \mathbb{R}^{512}$ are broadcast spatially across the 16×16 feature map. The γ -MLP is initialized with weight = 0 and bias = 1 (identity scale); the β -MLP with weight = 0 and bias = 0 (zero shift). This identity initialization ensures FiLM exerts no effect at epoch 0, training as a standard U-Net before quality-specific divergence emerges.

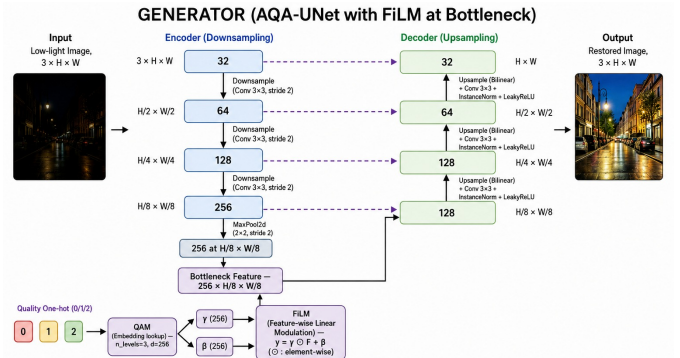
Decoder. Four upsampling stages apply 2×2 bilinear up-sampling, concatenate the corresponding encoder skip connection, and pass through a convolutional fusion block (Table II).

TABLE II
DECODER STAGE DIMENSIONS

Stage	Output Channels	Spatial Resolution
D4	256	32×32
D3	128	64×64
D2	64	128×128
D1	32	256×256

Output Layer. A 1×1 convolution projects the final 32-channel feature map to 3 output channels followed by Tanh activation:

$$\hat{\mathbf{y}} = \tanh(\text{Conv}_{1 \times 1}(d_1)) \in [-1, 1]^{3 \times 256 \times 256} \quad (8)$$



D. Discriminator Architecture

The discriminator D is a quality-conditioned 70×70 PatchGAN that outputs a 30×30 patch-level realism map, with each element summarizing the realism of a 70×70 receptive

field region. Six convolutional blocks with 4×4 kernels and LeakyReLU(0.2) activations form the feature extraction backbone (Table III).

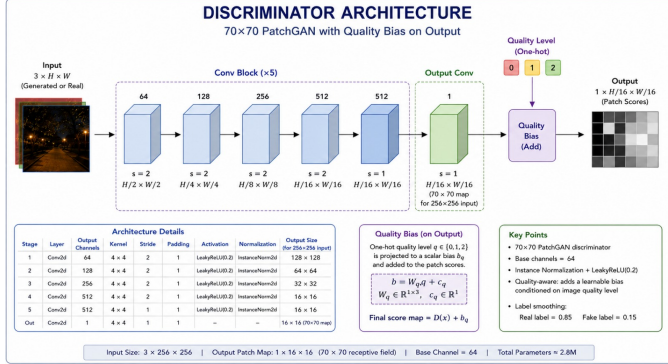
TABLE III
DISCRIMINATOR ARCHITECTURE

Layer	Out Ch.	Stride	Norm	Spatial Size
Input	—	—	—	256×256
C1	64	2	None	128×128
C2	128	2	BatchNorm	64×64
C3	256	2	BatchNorm	32×32
C4	512	2	BatchNorm	16×16
C5	512	2	BatchNorm	8×8
C6	512	1	BatchNorm	4×4
Out	1	1	None	30×30

Quality-Aware Bias Addition. The quality level q is mapped through a learned embedding ($d_q=16$), then projected to a scalar bias $b_q \in \mathbb{R}$ via Linear(16, 1). This bias is broadcast and added element-wise to the raw 30×30 score map:

$$S_{\text{final}}(i, j) = S_{\text{raw}}(i, j) + b_q \quad (9)$$

For severe inputs ($q=0$), b_q adjusts tolerance to residual artifacts. For mild inputs ($q=2$), b_q enforces stricter naturalness evaluation. The bias is initialized to zero. The discriminator has approximately 2.7M parameters (base channels $c=64$).



E. Loss Functions

The generator is optimized through:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{char}} \mathcal{L}_{\text{char}} + \lambda_{\text{ssim}} \mathcal{L}_{\text{ssim}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}} \quad (10)$$

with weights $\lambda_{\text{adv}}=0.5$, $\lambda_{\text{char}}=100.0$, $\lambda_{\text{ssim}}=5.0$, $\lambda_{\text{perc}}=0.05$.

1. Adversarial Loss — Least Squares GAN (LSGAN). Label smoothing with real target $a=0.85$ and fake target $b=0.15$:

$$\mathcal{L}_{\text{adv}}^D = \frac{1}{2} \mathbb{E}_{\mathbf{y}, q} [(D(\mathbf{y}, q) - a)^2] + \frac{1}{2} \mathbb{E}_{\hat{\mathbf{y}}, q} [(D(\hat{\mathbf{y}}, q) - b)^2] \quad (11)$$

$$\mathcal{L}_{\text{adv}}^G = \frac{1}{2} \mathbb{E}_{\hat{\mathbf{y}}, q} [(D(\hat{\mathbf{y}}, q) - a)^2] \quad (12)$$

2. Charbonnier Loss. A differentiable approximation of the L1 norm with $\varepsilon=10^{-3}$:

$$\mathcal{L}_{\text{char}} = \frac{1}{N} \sum_{i=1}^N \sqrt{(\hat{y}_i - y_i)^2 + \varepsilon^2} \quad (13)$$

3. SSIM Loss.

$$\mathcal{L}_{\text{ssim}} = 1 - \frac{1}{M} \sum_{\omega} \text{SSIM}_{\omega}(\hat{\mathbf{y}}, \mathbf{y}) \quad (14)$$

where:

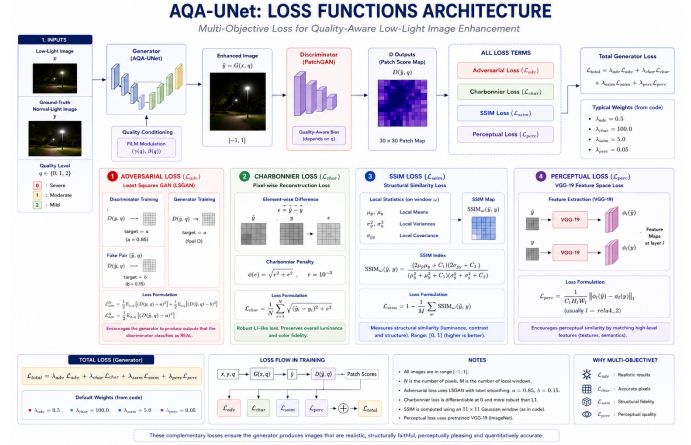
$$\text{SSIM}_{\omega}(\hat{\mathbf{y}}, \mathbf{y}) = \frac{(2\mu_{\hat{\mathbf{y}}}\mu_{\mathbf{y}} + C_1)(2\sigma_{\hat{\mathbf{y}}\mathbf{y}} + C_2)}{(\mu_{\hat{\mathbf{y}}}^2 + \mu_{\mathbf{y}}^2 + C_1)(\sigma_{\hat{\mathbf{y}}}^2 + \sigma_{\mathbf{y}}^2 + C_2)} \quad (15)$$

with $C_1 = (0.01)^2$ and $C_2 = (0.03)^2$.

4. Perceptual Loss — VGG-19. Feature matching at layer relu4_2:

$$\mathcal{L}_{\text{perc}} = \frac{1}{C_l H_l W_l} \|\phi_l(\hat{\mathbf{y}}) - \phi_l(\mathbf{y})\|_1 \quad (16)$$

Discriminator Stabilization. An adaptive discriminator skip mechanism pauses discriminator updates when $D(\text{real}) > 0.80$ and $D(\text{fake}) < 0.20$ simultaneously, preventing discriminator dominance.



V. DATASETS AND EXPERIMENTAL SETUP

A. Datasets

LOL-v2 Real contains paired low-light and normal-light images captured in real nighttime environments. Each pair is spatially aligned and represents the same scene, enabling direct evaluation using full-reference metrics such as PSNR and SSIM.

LSRW (our485 subset) includes approximately 485 paired low/normal-light image pairs collected from indoor and outdoor nighttime scenes with different illumination conditions and noise characteristics.

Combined Training Corpus. The LOL-v2 Real training split and the complete LSRW dataset are combined into a single corpus and divided into 80% training and 20% validation subsets using a fixed random seed (seed = 42). The LOL-v2 Real test split is reserved exclusively for final evaluation.

B. Preprocessing and Augmentation

All preprocessing and augmentation operations are applied identically to both low-light and corresponding normal-light image pairs to maintain spatial alignment.

Training pipeline:

- Resize to 320×320 using bicubic interpolation
- Random crop to 256×256
- Random horizontal flip with probability $p = 0.5$
- Random vertical flip with probability $p = 0.3$
- Random rotation within $\pm 15^\circ$ using bilinear interpolation
- Color jitter with brightness ± 0.10 , contrast ± 0.10 , and saturation ± 0.08
- Conversion to tensor followed by normalization to $[-1, 1]$ using $\mu = [0.5, 0.5, 0.5]$ and $\sigma = [0.5, 0.5, 0.5]$

Within-batch MixUp augmentation is applied during training with $\alpha = 0.4$, where the mixing coefficient follows $\lambda \sim \text{Beta}(0.4, 0.4)$.

Evaluation pipeline: images are resized to 256×256 followed by center cropping to 256×256 with the same normalization procedure. No random augmentation is applied during evaluation.

VI. RESULTS AND ABLATION STUDY

A. Quantitative Results

Table IV compares the proposed method with six baseline approaches, including classical enhancement methods, shallow learning models, CNN-based approaches, and GAN-based networks. Performance is evaluated using PSNR and SSIM.

TABLE IV
QUANTITATIVE COMPARISON ON COMBINED LOL-V2 REAL AND LSRW BENCHMARK

Method	PSNR (dB)	SSIM
Retinex [2]	15.8	0.48
Wavelet Denoising	17.3	0.54
Random Forest	18.6	0.59
SRCNN	19.8	0.63
U-Net [6]	20.4	0.69
SRGAN [10]	20.8	0.71
AQA-GAN (Ours)	22.6	0.82

AQA-GAN achieves a PSNR of 22.6 dB and an SSIM score of 0.82, exceeding the performance of SRGAN by 1.8 dB in PSNR and 0.11 in SSIM.

B. Ablation Study

Four model configurations are evaluated to examine the contribution of individual components in the proposed framework.

Baseline (Pixel losses only, no GAN). Training the U-Net generator using only Charbonnier and SSIM losses produces relatively high PSNR values but results in smoother textures and reduced perceptual detail. Without adversarial supervision, the model lacks a mechanism to encourage realistic texture reconstruction.

Effect of the GAN module. Introducing the quality-conditioned PatchGAN discriminator while disabling quality

conditioning ($q = 0$ for all samples) improves SSIM and restores sharper textures. The 70×70 PatchGAN provides localized adversarial supervision that improves perceptual quality.

Effect of quality conditioning (FiLM). Activating FiLM-based conditioning improves enhancement quality for severely degraded inputs ($q = 0$). As training progresses, PSNR differences between quality levels become more distinct, indicating that the FiLM embeddings learn degradation-dependent feature transformations rather than remaining near identity mappings.

Effect of the quality-aware discriminator bias. Incorporating the learnable quality bias term b_q improves training stability and reduces over-enhancement in moderately illuminated images. Without this bias, identical adversarial constraints are applied to all inputs regardless of degradation severity. The adaptive discriminator skip mechanism further prevents discriminator dominance during later training stages.

Overall, the ablation results indicate that each component contributes independently to the final performance. The GAN module improves perceptual realism, FiLM conditioning enables degradation-adaptive enhancement, and the quality-aware discriminator provides balanced adversarial supervision across varying illumination conditions.

VII. FUTURE WORK

Continuous Quality Conditioning. The current Quality Assessment Module (QAM) assigns images to three discrete degradation levels. Since illumination degradation varies continuously in real-world environments, future work may replace discrete labels with continuous quality representations. Diffusion-based restoration models such as Palette [15] provide a suitable framework for integrating continuous quality guidance into the restoration process.

Real-Time Deployment on Edge Devices. The current AQA-GAN generator contains approximately 3.6 million parameters and is not optimized for low-latency inference. Techniques such as structured pruning, knowledge distillation, and quantization-aware training may improve deployment efficiency on embedded platforms and mobile imaging systems.

Video Low-Light Enhancement. The present framework operates on individual image pairs and does not explicitly model temporal consistency. Frame-wise enhancement may introduce flickering and color instability in video sequences. Prior work by Zhang et al. [16] demonstrated that temporal consistency can improve low-light video enhancement. Extending AQA-GAN with recurrent or attention-based temporal propagation is a potential direction for future research.

Multi-Task Learning. Future extensions may jointly optimize enhancement, denoising, deblurring, and super-resolution within a unified framework using a shared encoder conditioned on both degradation level and task identity.

Integration with Diffusion-Based Restoration. The proposed quality-aware conditioning strategy may also be incorporated into diffusion-based restoration frameworks by conditioning the reverse diffusion process on estimated degradation severity.

VIII. CONCLUSION

This work introduced AQA-GAN, an adaptive quality-aware generative adversarial framework for nighttime low-light image enhancement. The proposed method addresses the limitations of uniform enhancement by incorporating quality-aware conditioning throughout the enhancement pipeline. A lightweight Quality Assessment Module estimates image degradation using brightness, contrast, and local texture statistics, assigning each input to one of three degradation levels without requiring additional annotations or trainable parameters.

The estimated quality label conditions both the FiLM-based U-Net generator and the 70×70 PatchGAN discriminator. The generator applies degradation-aware affine modulation at the bottleneck feature level, while the discriminator adjusts adversarial supervision using a learnable quality-dependent bias term. Training stability is improved using a composite objective consisting of Charbonnier, SSIM, perceptual, and adversarial losses together with MixUp augmentation, label smoothing, and adaptive discriminator skipping.

Experiments conducted on the combined LOL-v2 Real and LSRW benchmark demonstrate that AQA-GAN achieves 22.6dB PSNR and 0.82 SSIM, outperforming all evaluated baselines. Improvements of 1.8dB in PSNR and 0.11 in SSIM are observed compared with the strongest GAN-based baseline. Performance gains across different degradation levels further indicate that the conditioning mechanism adapts enhancement strength according to input quality.

The proposed quality-aware conditioning strategy is independent of the underlying restoration architecture and can be extended to related image restoration tasks such as denoising, dehazing, deblurring, and super-resolution. The framework provides a basis for future research in adaptive and degradation-aware image restoration methods.

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